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ASSIGNMENT [CSE 37]
FACTORS AFFECTING OCEAN POLLUTION

SUBMITTED TO,
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Factors Affecting Ocean Pollution

Introduction

Ocean pollution is driven by five broad groups of factors: land-based sources, plastic and solid waste, maritime and offshore activities, atmospheric and climate-related inputs, and human governance and social conditions. Together they decide how much harmful material reaches the sea, how long it stays, and how badly it damages marine life.

The ocean covers about 70% of Earth's surface, produces a large share of the oxygen we breathe, feeds billions of people, and regulates climate. Because water moves continuously through currents, tides and rivers, a pollutant released in one place rarely stays there. Fertiliser from a farm inland, a plastic bottle dropped in a city drain, and smoke from a factory chimney can all end up in the same stretch of open water.

This document explains each of the five factors in turn. Every factor is broken into sub-points that describe the source, how the pollution reaches the ocean, and the effect it has on ecosystems, human health and economies. The final section shows how the factors interact and what can be done.

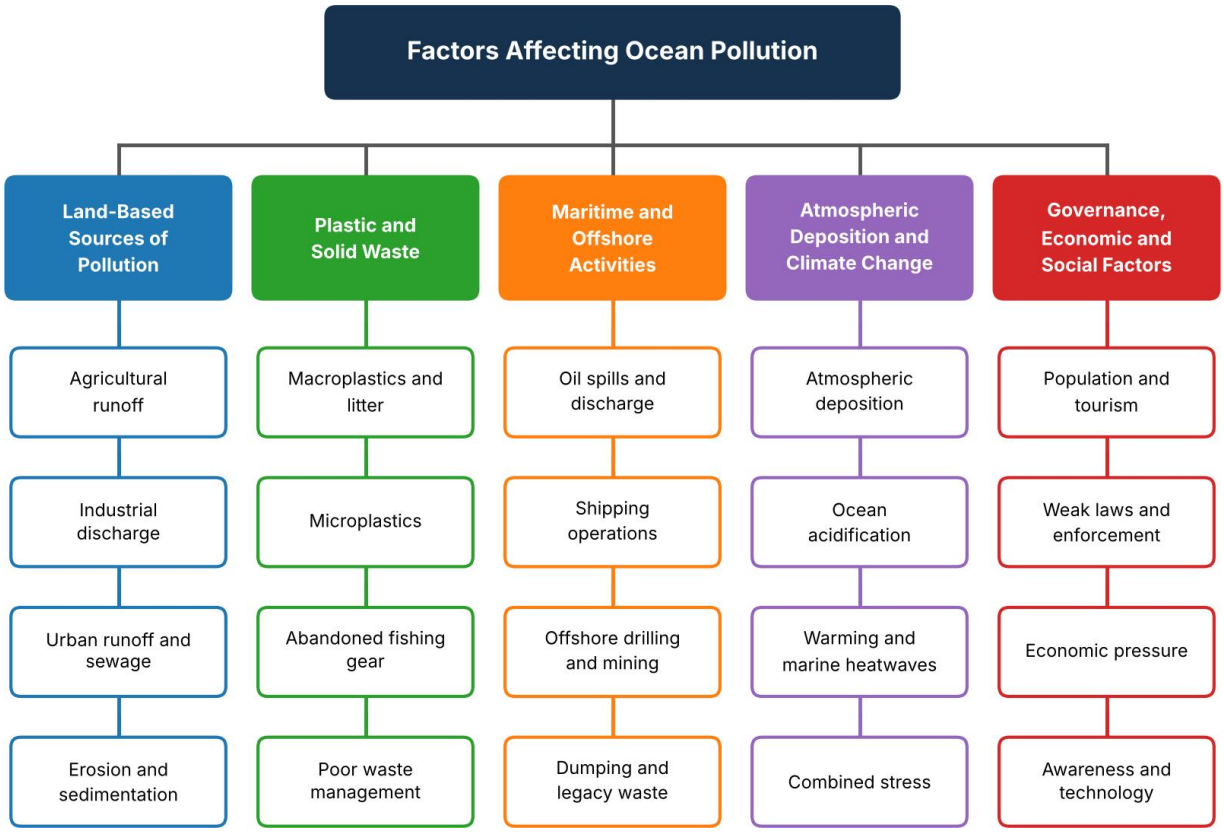


Figure 1: Main factors affecting ocean pollution and their sub-factors

1. Land-Based Sources of Pollution

Around 80% of marine pollution is commonly estimated to begin on land, carried to the sea by rivers, drains and groundwater.

- **Agricultural runoff:** Fertiliser nitrogen and phosphorus trigger algal blooms. When the algae die, decomposition uses up oxygen and creates dead zones, such as the one in the Gulf of Mexico. Pesticides and livestock waste add toxins and bacteria.
- **Industrial discharge:** Factories and mines release heavy metals such as mercury and lead, along with persistent chemicals that build up in fish. Power plants also return warm cooling water that stresses corals.
- **Urban runoff and sewage:** Untreated sewage spreads disease and consumes oxygen, while storm drains wash oil, litter and metals straight to the coast. Medicines and cosmetic chemicals pass through treatment plants only partly removed.
- **Erosion and sedimentation:** Deforestation, construction and dredging send soil into the sea. The cloudy water blocks sunlight for seagrass and coral, and the silt smothers bottom-dwelling life.

2. Plastic and Solid Waste

Plastic is the most visible and persistent pollutant, with an often-cited estimate of roughly 8 million tonnes entering the ocean each year (published figures vary).

- **Macroplastics and litter:** Bottles, bags and packaging reach the sea through rivers and storm drains. Turtles, seabirds and whales swallow them or become entangled, and currents gather them into gyres such as the Great Pacific Garbage Patch.
- **Microplastics:** Fragments under 5 mm come from broken-down litter, microbeads, synthetic clothing fibres and tyre wear. They enter plankton, shellfish and fish, carry toxic chemicals, and have been found from polar ice to the deep seabed.
- **Abandoned fishing gear:** Lost nets, lines and traps keep catching fish, turtles and mammals for years ("ghost fishing") and damage coral reefs.
- **Poor waste management:** Where collection and recycling are limited, waste is dumped or burned and ends up in waterways. Growing demand for single-use packaging increases the volume that must be handled.

3. Maritime and Offshore Activities

Shipping and offshore industry put pollutants directly into the sea. Their total volume is smaller than land-based inputs, but the effects can be sudden and severe.

- **Oil spills and discharge:** Accidents such as Exxon Valdez (1989) and Deepwater Horizon (2010) coated coasts and killed wildlife. Smaller, repeated releases from tank washing and engine leaks add a steady oil load.

- **Shipping operations:** Ballast water moves invasive species between ports, toxic antifouling paints have harmed shellfish, and engine exhaust, sewage, garbage and noise affect water quality and marine animals.
- **Offshore drilling and mining:** Drilling muds, produced water, seabed mining plumes and seismic surveys release contaminants, smother organisms and disturb marine mammals.
- **Dumping and legacy waste:** Contaminated dredged material is sometimes disposed of at sea, and past dumping of chemical munitions and industrial waste, plus wrecks, can leak for decades.

4. Atmospheric Deposition and Climate Change

The sea absorbs gases and particles from the air and responds to global warming, which changes its chemistry and temperature.

- **Atmospheric deposition:** Smoke and dust from industry, vehicles and burning carry metals, nitrogen and chemicals onto the sea surface. Mercury from coal burning travels far before falling into the ocean and accumulating in fish.
- **Ocean acidification:** The ocean absorbs roughly a quarter of human carbon dioxide emissions. Surface pH has dropped about 0.1 units since the industrial era (around 30% more acidic), which makes it harder for corals, oysters and plankton to build shells.
- **Warming and marine heatwaves:** Heat causes coral bleaching and pushes fish toward cooler water. Warmer water also holds less oxygen, which worsens dead zones.
- **Combined stress:** Rising seas and stronger storms carry more waste into the ocean, and warm, acidic, low-oxygen water leaves marine life less able to cope with toxins and disease.

5. Governance, Economic and Social Factors

Laws, income levels and habits decide how much pollution is produced and whether it is controlled, which is why similar coastlines can have very different outcomes.

- **Population and tourism:** Many people live near the coast, and growing cities, hotels and cruise ships increase sewage, waste and runoff, especially in peak seasons.
- **Weak laws and enforcement:** Some countries have limited standards, or lack inspectors and penalties, so illegal dumping continues. The high seas have no single owner, and treaties such as MARPOL depend on states carrying them out.
- **Economic pressure:** Sewage plants and waste collection are costly for poorer municipalities, and some industries choose the cheapest disposal. Poorer coastal communities suffer most from contaminated seafood and damaged fisheries.
- **Awareness and technology:** Littering and single-use plastic persist when people do not link daily habits to ocean health. Satellites, sensors and cleaner technology help, but only if adopted and monitored.

By applying the Fuzzy-AHP Algorithm

The Fuzzy Analytic Hierarchy Process (Fuzzy-AHP) is used to find which factors affect ocean pollution most. Expert judgements are written as triangular fuzzy numbers (l, m, u) instead of single crisp values, so the vagueness of human opinion is kept in the calculation. The method used is Buckley's fuzzy geometric mean method.

Main Factors of Ocean Pollution:

- **F1:** Land-Based Sources of Pollution
- **F2:** Plastic & Solid Waste
- **F3:** Maritime & Offshore Activities
- **F4:** Atmospheric Deposition & Climate Change
- **F5:** Governance, Economic & Social Factors

Fuzzy Pairwise Comparison Matrix (Main Factors)

	F1	F2	F3	F4	F5	Global Weight
F1	1, 1, 1	1.00, 1.50, 2.00	2.00, 2.50, 3.00	1.50, 2.00, 2.50	1.00, 1.30, 1.80	0.29469
F2	0.50, 0.67, 1.00	1, 1, 1	1.50, 2.00, 2.50	1.20, 1.60, 2.10	0.70, 1.00, 1.40	0.22196
F3	0.33, 0.40, 0.50	0.40, 0.50, 0.67	1, 1, 1	0.70, 1.00, 1.40	0.40, 0.60, 0.90	0.12525
F4	0.40, 0.50, 0.67	0.48, 0.62, 0.83	0.71, 1.00, 1.43	1, 1, 1	0.50, 0.70, 1.00	0.14169
F5	0.56, 0.77, 1.00	0.71, 1.00, 1.43	1.11, 1.67, 2.50	1.00, 1.43, 2.00	1, 1, 1	0.21640

Consistency check: $\lambda max = 5.1419$, CI = 0.0355, CR = 0.0317 (< 0.1, consistent).

1. Land-Based Sources of Pollution (sub-factors)

- **S1:** Agricultural runoff
- **S2:** Industrial discharge
- **S3:** Urban runoff and sewage
- **S4:** Erosion and sedimentation

Fuzzy Pairwise Comparison Matrix

	S1	S2	S3	S4	Normalized Weight
S1	1, 1, 1	1.00, 1.40, 1.90	0.80, 1.10, 1.50	2.00, 2.60, 3.20	0.33034
S2	0.53, 0.71, 1.00	1, 1, 1	0.60, 0.90, 1.30	1.50, 2.00, 2.50	0.25081
S3	0.67, 0.91, 1.25	0.77, 1.11, 1.67	1, 1, 1	1.80, 2.30, 2.80	0.29183
S4	0.31, 0.38, 0.50	0.40, 0.50, 0.67	0.36, 0.43, 0.56	1, 1, 1	0.12702

Consistency check: $\lambda max = 4.0890$, CI = 0.0297, CR = 0.0330 (< 0.1, consistent).

2. Plastic & Solid Waste (sub-factors)

- **S5:** Macroplastics and litter
- **S6:** Microplastics
- **S7:** Abandoned fishing gear
- **S8:** Poor waste management

Fuzzy Pairwise Comparison Matrix

	S5	S6	S7	S8	Normalized Weight
S5	1, 1, 1	0.50, 0.70, 1.00	1.80, 2.30, 2.80	0.80, 1.10, 1.50	0.26868
S6	1.00, 1.43, 2.00	1, 1, 1	2.20, 2.80, 3.40	1.20, 1.60, 2.10	0.36805
S7	0.36, 0.43, 0.56	0.29, 0.36, 0.45	1, 1, 1	0.40, 0.50, 0.70	0.12410
S8	0.67, 0.91, 1.25	0.48, 0.62, 0.83	1.43, 2.00, 2.50	1, 1, 1	0.23918

Consistency check: $\lambda max = 4.0810$, CI = 0.0270, CR = 0.0300 (< 0.1, consistent).

3. Maritime & Offshore Activities (sub-factors)

- **S9:** Oil spills and discharge
- **S10:** Shipping operations
- **S11:** Offshore drilling and mining
- **S12:** Dumping and legacy waste

Fuzzy Pairwise Comparison Matrix

	S9	S10	S11	S12	Normalized Weight
S9	1, 1, 1	1.20, 1.60, 2.10	1.80, 2.20, 2.70	2.40, 3.00, 3.60	0.40911
S10	0.48, 0.62, 0.83	1, 1, 1	1.00, 1.50, 2.00	1.60, 2.10, 2.60	0.26930
S11	0.37, 0.45, 0.56	0.50, 0.67, 1.00	1, 1, 1	0.90, 1.30, 1.80	0.18428
S12	0.28, 0.33, 0.42	0.38, 0.48, 0.62	0.56, 0.77, 1.11	1, 1, 1	0.13731

Consistency check: $\lambda max = 4.0790$, CI = 0.0263, CR = 0.0293 (< 0.1, consistent).

4. Atmospheric Deposition & Climate Change (sub-factors)

- **S13:** Atmospheric deposition
- **S14:** Ocean acidification
- **S15:** Warming and marine heatwaves

- **S16:** Combined stress

Fuzzy Pairwise Comparison Matrix

	S13	S14	S15	S16	Normalized Weight
S13	1, 1, 1	0.40, 0.50, 0.70	0.50, 0.70, 1.00	1.20, 1.50, 2.00	0.20524
S14	1.43, 2.00, 2.50	1, 1, 1	1.10, 1.50, 2.00	1.80, 2.30, 2.80	0.37608
S15	1.00, 1.43, 2.00	0.50, 0.67, 0.91	1, 1, 1	1.30, 1.70, 2.20	0.26785
S16	0.50, 0.67, 0.83	0.36, 0.43, 0.56	0.45, 0.59, 0.77	1, 1, 1	0.15083

Consistency check: $\lambda max = 4.0901$, CI = 0.0300, CR = 0.0334 (< 0.1, consistent).

5. Governance, Economic & Social Factors (sub-factors)

- **S17:** Population and tourism
- **S18:** Weak laws and enforcement
- **S19:** Economic pressure
- **S20:** Awareness and technology

Fuzzy Pairwise Comparison Matrix

	S17	S18	S19	S20	Normalized Weight
S17	1, 1, 1	0.40, 0.60, 0.90	0.90, 1.30, 1.80	1.50, 2.00, 2.50	0.25650
S18	1.11, 1.67, 2.50	1, 1, 1	1.80, 2.40, 3.00	2.20, 2.80, 3.40	0.41574
S19	0.56, 0.77, 1.11	0.33, 0.42, 0.56	1, 1, 1	1.00, 1.40, 1.90	0.18989
S20	0.40, 0.50, 0.67	0.29, 0.36, 0.45	0.53, 0.71, 1.00	1, 1, 1	0.13786

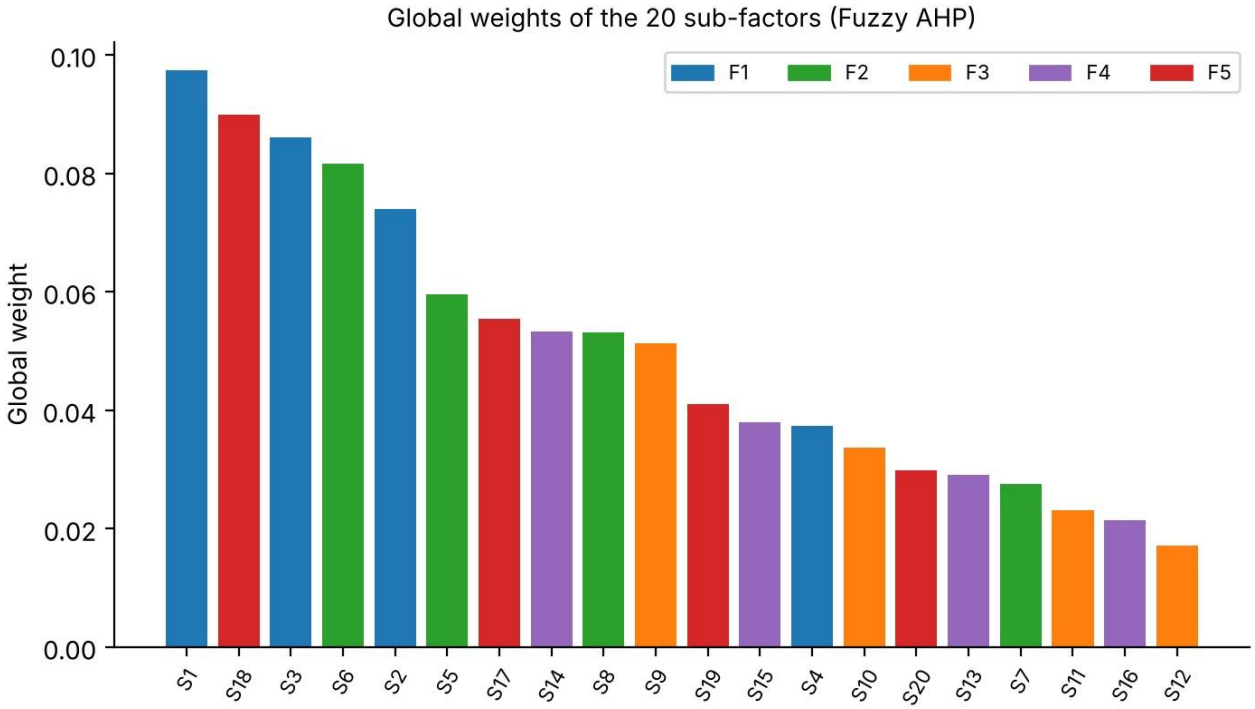
Consistency check: $\lambda max = 4.1005$, CI = 0.0335, CR = 0.0372 (< 0.1, consistent).

Overall weight calculation

Main Factor	Global Weight (Wi)	Sub-factor	Normalized Weight (wij)	Wi × wij
F1 – Land-Based Sources of Pollution	0.29469	S1	0.33034	0.09735
		S2	0.25081	0.07391
		S3	0.29183	0.08600
		S4	0.12702	0.03743
F2 – Plastic & Solid Waste	0.22196	S5	0.26868	0.05964
		S6	0.36805	0.08169
		S7	0.12410	0.02754
		S8	0.23918	0.05309
F3 – Maritime & Offshore Activities	0.12525	S9	0.40911	0.05124
		S10	0.26930	0.03373
		S11	0.18428	0.02308
		S12	0.13731	0.01720
F4 – Atmospheric Deposition & Climate Change	0.14169	S13	0.20524	0.02908
		S14	0.37608	0.05329
		S15	0.26785	0.03795
		S16	0.15083	0.02137
F5 – Governance, Economic & Social Factors	0.21640	S17	0.25650	0.05551
		S18	0.41574	0.08997
		S19	0.18989	0.04109
		S20	0.13786	0.02983

Check: sum of all global weights = 1.00000.

Graphical Representation of Global Weights



Relative global weights of the 20 sub-factors of ocean pollution calculated using Fuzzy-AHP, ranked from highest to lowest.

Results & Discussion:

- Among the five main factors, F1 (Land-Based Sources of Pollution) has the highest weight of 0.29469, followed by F2 (Plastic & Solid Waste) with 0.22196 and F5 (Governance, Economic & Social Factors) with 0.21640. F4 (Atmospheric Deposition & Climate Change) has 0.14169 and F3 (Maritime & Offshore Activities) has the lowest weight, 0.12525.
- Among the 20 sub-factors, S1 (Agricultural runoff) has the highest global weight of 0.09735, followed by S18 (Weak laws and enforcement, 0.08997) and S3 (Urban runoff and sewage, 0.08600). The top five sub-factors together account for 42.9% of the total weight.
- Land-based inputs and plastic waste together carry more than half of the total weight, so controlling what leaves the land and how waste is handled gives the largest reduction in ocean pollution.
- Weak laws and enforcement is the most important governance sub-factor, which shows that regulations only reduce pollution when they are actually enforced.
- All consistency ratios are below 0.1, so the expert judgements used in the matrices are acceptable.

Conclusion:

This study used Fuzzy-AHP to rank the factors behind ocean pollution. Land-based sources and plastic and solid waste are the highest-priority factors, with agricultural runoff, urban sewage and runoff, and weak law enforcement among the most important sub-factors. Preventing pollution at its source on land, cutting plastic waste, and enforcing existing rules should therefore receive first priority in ocean protection plans, while shipping controls and climate action continue in support. Fuzzy-AHP gives a systematic, transparent way to set these priorities and to allocate limited resources.